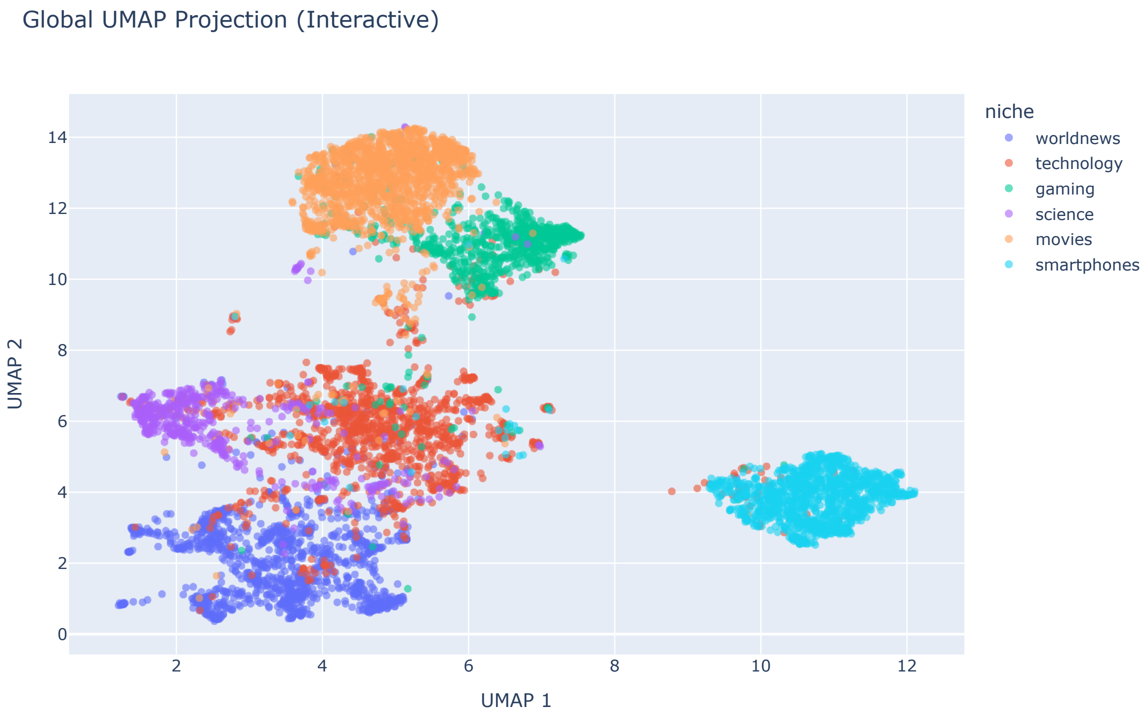
```
import umap
import plotly.express as px

# Prepare data for global UMAP
X_all = np.vstack(final_df['embedding'].values)
reducer_all = umap.UMAP(n_components=2, random_state=42)
X_all_2d = reducer_all.fit_transform(X_all)

umap_df = pd.DataFrame(X_all_2d, columns=['UMAP 1', 'UMAP 2'])
umap_df['niche'] = final_df['niche']
umap_df['title'] = final_df['title']

fig = px.scatter(
    umap_df, x='UMAP 1', y='UMAP 2', color='niche',
    hover_data=['title'], opacity=0.6,
    title='Global UMAP Projection (Interactive)',
    width=900, height=600
)
fig.show()
```

```
/usr/local/lib/python3.12/dist-packages/umap/umap_.py:1952: UserWarning: n_jobs value 1 overridden to 1 by setting random_st
  warn(
```




```
import plotly.express as px
import plotly.graph_objects as go
import umap
import numpy as np

def plot_niche_outliers(niche_name, df):
    subset = df[df['niche'] == niche_name].copy()
    X = np.vstack(subset['embedding'].values)

    # Use UMAP for 2D projection
    reducer = umap.UMAP(n_components=2, random_state=42, n_neighbors=15, min_dist=0.1)
    X_2d = reducer.fit_transform(X)

    subset['umap_1'] = X_2d[:, 0]
    subset['umap_2'] = X_2d[:, 1]

    # Separate normal points and outliers
    normal_pts = subset[~subset['outlier']]
    outlier_pts = subset[subset['outlier']]

    # Base plot for normal points
    fig = px.scatter(
        normal_pts, x='umap_1', y='umap_2',
        color=normal_pts['cluster'].astype(str),
        hover_data=['title'],
        opacity=0.4,
        title=f"Semantic Map & Outliers: {niche_name}",
```